TEXORA · LECTURE SHEET

Mechanical Properties

The tensile story of fibres: the specific-stress/strain curve and how to read it — tenacity, breaking extension, initial modulus, yield, work of rupture — plus elastic recovery, creep and the bending and abrasion behaviours that decide wear life in service.

TE-3209 · TEXTILE PHYSICS · TOPIC 3 · LEVEL 3 TERM 2

1. The tensile curve in textile units

Fibres are tested in tension and the results plotted as **specific stress** (load per linear density, cN/tex — equal to tenacity at the break point) against **strain** (%). Textile units free the comparison from fineness: a thick and a fine fibre of the same polymer trace the same curve. The vocabulary of the curve:

- **Initial modulus** — slope at the origin: stiffness, the “hand” and drape contributor (high for flax/glass, low for wool/elastane).
- **Yield point** — where easy, largely recoverable straining gives way to permanent set (wool’s long yield region is its comfort secret).
- **Tenacity** — specific stress at break; **breaking extension** — strain at break.
- **Work of rupture** — area under the whole curve: energy absorbed before failure, the toughest single predictor of wear and impact survival.

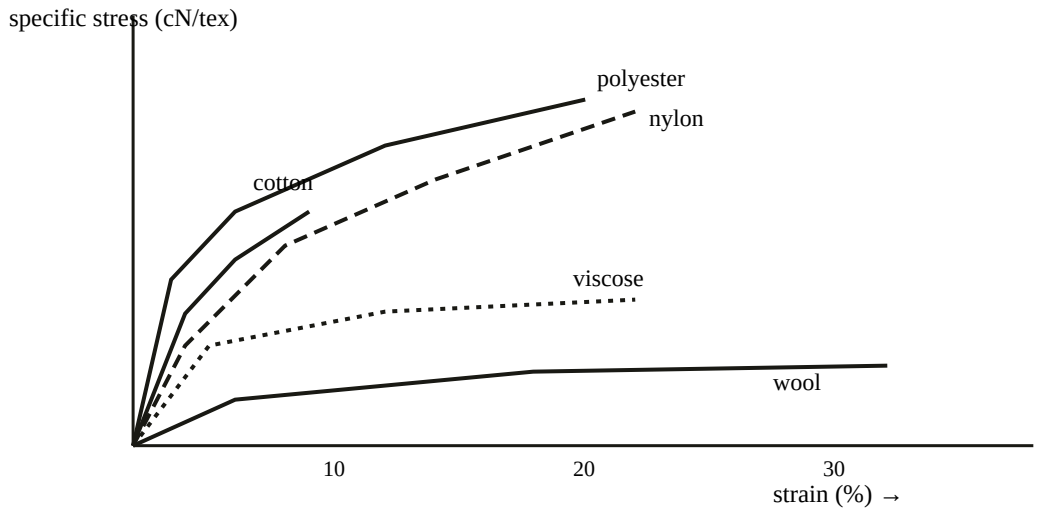


Figure 1 — Characteristic tensile curves of the five fibres (schematic, from typical values): cotton steep and short, wool low and long, viscose weak, nylon and polyester strong with polyester the stiffer start.

WORKED EXAMPLE — TENACITY AND MODULUS FROM A TEST

A polyester filament of 1.7 dtex breaks at 7.65 cN. Linear density = 0.17 tex, so tenacity = 7.65/0.17 = **45 cN/tex** (a high-tenacity grade). On its curve, at 1% strain the specific stress is 10 cN/tex, so initial modulus = 10/0.01 = 1000 cN/tex = 10 N/tex. Converting to engineering modulus: E(GPa) = modulus (N/tex) × density (g/cm³) = 10 × 1.38 = **13.8 GPa** — the bridge between textile and engineering units, valid for any fibre.

2. Time-dependent behaviour — recovery, creep, relaxation

Fibres are viscoelastic: response depends on how long and how fast. **Elastic recovery** measures how much imposed strain returns on unloading — wool and nylon recover nearly fully from small strains (wrinkle resistance, carpet resilience), viscose poorly (its garments bag). **Creep** is slow continued extension under constant load (a hanging curtain lengthens; a rope under load keeps giving); **stress relaxation** is the mirror — held at fixed extension, the load decays (a stretched waistband loosens). Recovery falls as the strain, duration or temperature rise; moisture plasticises and deepens all time effects in hydrophilic fibres. Abrasion and flex life close the practical list: nylon's flex-abrasion resistance is the best of the apparel fibres — the reason it reinforces heels and toes at blend fractions under 20%.

Temperature writes the last line: approaching Tg, amorphous chains mobilise and modulus collapses (the basis of heat-setting, Topic 7); below it, fibres stiffen and embrittle. Every mechanical number in this sheet is understood to be at standard atmosphere — 20 °C, 65% RH — unless stated.

3. Key points

- Plot specific stress vs strain: tenacity, breaking extension, initial modulus, yield and work of rupture are the five readings.
- Tenacity example: 7.65 cN on 0.17 tex = 45 cN/tex; modulus converts by E(GPa) = (N/tex) × density → 13.8 GPa.
- Work of rupture (curve area) predicts survival better than tenacity alone — wool's long curve compensates its low peak.
- Viscoelasticity: recovery, creep and relaxation — moisture and temperature amplify all three.
- Nylon owns flex-abrasion life; viscose owns the weakest recovery; conditions of test (atmosphere, strain rate) are part of every quoted number.

4. Practice

Answer in your own words; brief model answers follow.

1. A cotton fibre of 1.8 dtex breaks at 6.3 cN. Find its tenacity in cN/tex and in g/den.
2. Sketch-reading: fibre A reaches 30 cN/tex at 6% strain; fibre B reaches 18 cN/tex at 28% with a broad curve. Which has the higher work of rupture, approximately, and which suits a climbing rope vs a sewing thread?
3. A yarn creeps 0.5% in the first hour under load and 0.05% per hour thereafter. Find its total extension after 24 h on a 2 m length, and comment for a hanging fabric.
4. Why does wool's low initial modulus feel "soft" in fabric hand while flax feels harsh, although linen fabric is highly prized?
5. A filament shows initial modulus 5 N/tex and density 1.14 g/cm³. Convert to GPa and name the likely fibre.
6. Explain why a nylon stocking survives flexing that destroys a viscose one, in curve-and-recovery terms.

Answer key

1. Linear density = 0.18 tex; tenacity = 6.3/0.18 = 35 cN/tex. In g/den: 35/8.83 ≈ 3.96 ≈ 4.0 g/den — mid-range cotton.
2. Approximate areas (triangle): A ≈ ½ × 30 × 6 = 90 units; B ≈ ½ × 18 × 28 = 252 units — B's work of rupture is nearly three times A's. Climbing rope: B (energy absorption is survival). Sewing thread: A-type character (high tenacity at low extension gives firm seams and low stretch in the stitch).
3. Extension = 0.5 + 23 × 0.05 = 1.65% → 0.0165 × 2000 = 33 mm of growth in a day — visible drop in a curtain or banner, and the reason sustained-load applications specify low-creep fibres (polyester over nylon, both over viscose).
4. Initial modulus is the fabric's first resistance to bending and compression in the hand: wool's shallow opening slope bends easily, feeling soft and warm. Flax's steep slope feels crisp and harsh — prized as linen's cool, structured drape and crease-sharp tailoring, "harsh" being a context judgement, not a defect.
5. E = 5 × 1.14 = 5.7 GPa — nylon 6.6 (low density is the giveaway).
6. Nylon combines high work of rupture (a broad, high curve) with near-complete recovery from small repeated strains, so each flex is absorbed and returned; viscose has a lower curve, poor recovery and wet weakness, so flex damage accumulates cycle by cycle until filaments fibrillate and fail.